From lower degree to a testable solver question

DEC-20261029-266f95 — approve scoped archival acceptance

The frozen producer and one independent derivation agree on exact symbolic counts and a conditional complementary-point reference. **Seven review joints hold.** The software execution component remains **inconclusive in the independent review**; a separate root receipt records **18 fixed correctness tests passing**. No scientific or comparative run occurred. Archival effect requires the native ledger binding.

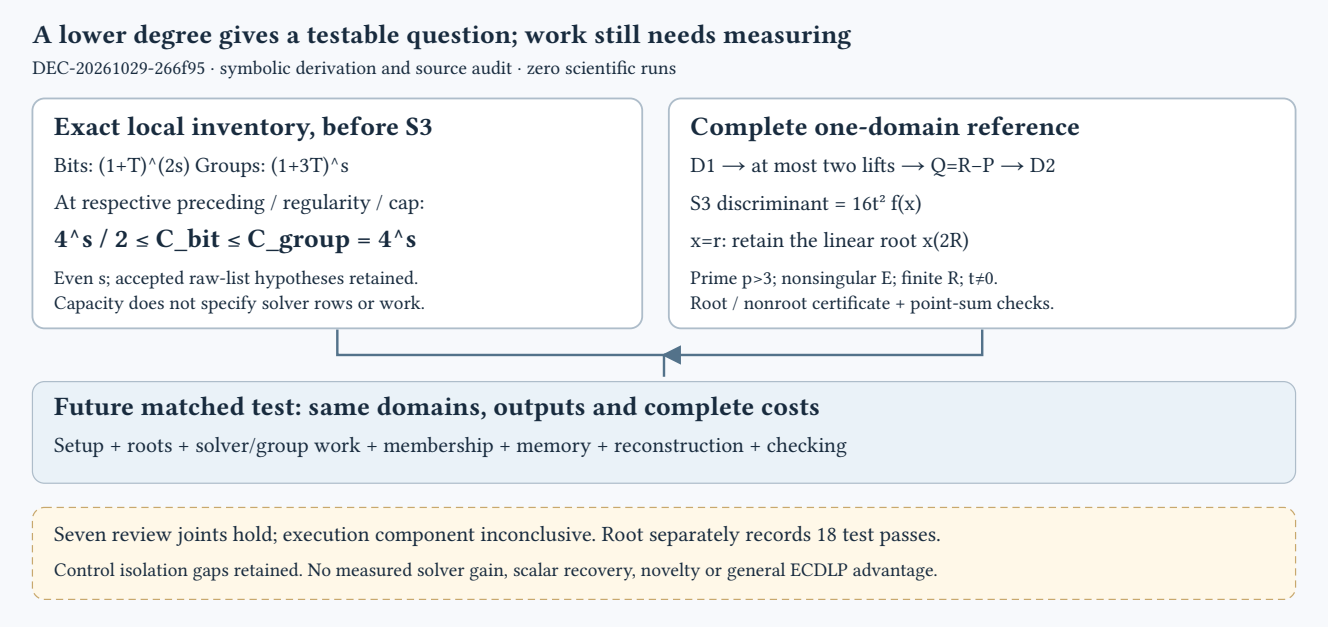
Observation: what the lower degree counts

For the accepted raw lists, retain even $s \geq 2$, $p > \max(3,s)$, nonzero original weights, two disjoint s-bit coordinate blocks, nonsingular short Weierstrass curve and finite rational $R=(r,t)$, $t \neq 0$. DEC-0857a9 supplies $d_{bit}=1+3s/2$ and $d_{group}=s+1$. These are prior premises.

Before S3, the local highest-part Hilbert series are $(1+T)^{2s}$ and $(1+3T)^{s}$. Prefixes are $C_{bit}(D)=\Sigma \binom{2s}{j}$, $j \leq \min(D,2s)$, and $C_{group}(D)=\Sigma \binom{s}{j}\cdot 3^j$, $j \leq \min(D,s)$.

Position	Bit cutoff	Group cutoff	Equality $C_{bit}=4^s$
One below	$3s/2$	s	Never for even $s \geq 2$
Regularity	$3s/2+1$	$s+1$	Only $s=2$
Mutant cap	$3s/2+2$	$s+2$	Only $s=2,4$

At every paired position, **$4^s/2 \leq C_{bit} \leq C_{group}=4^s$**. These inventories share an exponential order. Ambient capacities instead equal $\binom{2s+D}{D}$ and $\binom{3s+D}{D}$. Neither is necessary algorithm work. Adding constant one leaves the local inventory unchanged but makes the full system trivially inconsistent. Local and full highest-part quotients, affine quotients, and actual rows/pivots are distinct.



Symbolic formulas and a method diagram; no measured values, RUN IDs, samples or uncertainty bars. Producer Sections 1–4; independent pre-code Sections 1–5.

Comparison: complete one-domain reference

Assume any **mathematically prime p>3**, nonsingular E:$y^2=f(x)=x^3+ax+b$, finite R=(r,t), t≠0, and duplicate-free affine domains D1,D2. Domains may overlap. Enumerate all D1 lifts P, compute Q=R−P, exclude infinity, and test x(Q) in D2. All mode outputs the complete ordered projected set and witnesses; K≤2|D1|−1 when r∈D1, otherwise K≤2|D1|. First mode always declares completeness false.

With h=x−r and c=−rx−2r^2−a, the independent calculation gives:

$$S_3 = h^2z^2 + (2hc - 4t^2)z + c^2 - 4t^2(x + r)$$

$$\Delta = 16t^2(x^3 + ax + b).$$

For h≠0, $y^2=f(x)$ gives roots (−(2hc−4t^2)±4ty)/(2h^2), with one root when y=0. For h=0, retain the unique linear root x(2R). P=R gives infinity, while P=−R gives finite 2R. Tangencies, zero ordinates and order-three targets remain covered symbolically. No operational full-y scan or D1×D2 loop is used.

The separate checker imports no producer helper. It verifies exact D1 coverage, roots and both signs, complete second roots and projected-set equality, then point sums independently. For a nonroot, a global Euler-certified nonsquare Z and nonzero w with w^2=Zf(x) give a certificate. **Producing w also costs a square root.** Division-free chord/tangent equations check each witness. The API does not prove primality or execution counters.

Actual independent joint	Verdict
Prefix counts; ambient/work distinction	holds; holds
Rational completeness; complementary reference	holds; holds
Certificate checker; cost/setup; decision boundary	holds; holds; holds
Mode and software scope	Mode audit holds; execution inconclusive

The reviewer found no source defect falsifying the conditional algorithm/checker. It ran no tests and read no result receipt. Root separately observed one fixed suite of 18 passes at p=5,7,13, with two p=7 curves and tiny exhaustive software oracles. This does not independently replicate execution or prove all-prime software behavior.

Inference: the costs that a future test must pay

Let B1=|D1|. Producer cost is setup + O(B1·(E+S_actual+G+M+1)) + O(K log(K+1)) comparisons + certificate/output construction and serialization. E is Euler testing, S_actual includes every root call's initialization and verification, G is group work, and M is D2 membership. Nonsquare rows require twisted roots too. Deterministic nonsquare search is charged by actual probes; expected-constant search is not claimed. The recorded euler_probes field counts setup only.

Checker cost is its own setup + one Euler(Z) + O(B1·(I+M+1)) root/certificate work, output comparison and O(K) field operations for division-free witnesses. I is inversion; no new square root is extracted. Retained records, indexing and expected hash assumptions, sorting, input bit lengths, serialization and arithmetic bit costs remain charged. This is ordinary Tonelli–Shanks without a constant-time claim.

Future comparisons must match domains, projection multiplicity, first/all mode, exceptions, setup/reuse, actual solver work, memory, reconstruction and checking, including this reference. New source/protocol and native admission are required before any comparative run. The reviewer's sharper asymptotics stay outside this frozen producer acceptance.

Limitations and the concrete next step

The review retains three coverage limits: no dedicated mutation of a genuinely square radicand into a nonsquare-status row; invalid-input checker tests can also reject on a changed input digest; no dedicated order-three target fixture or all Tonelli–Shanks valuations. These limit control claims, without refuting the symbolic source argument. Before comparative use, isolate them in a prospective additive software amendment and preserve the frozen package.

The lower degree does not establish or rule out a SAT/Gröbner gain. Two solving-degree upper bounds do not order actual solving degrees. Arbitrary auxiliary or reduced encodings need their own graded proof. No formal proof, novelty finding, first/last-fall improvement, measured solver advantage, unknown-scalar recovery or general ECDLP speedup is established. GOAL-PFDR-fd09e0 remains active; no H/EV/KN or goal transition occurs. Historical PNEXT invalid_output remains unchanged.

Review and custody qualifications

Producer TASK-20261029-2e3655: `8c9cbfb9c785`; verified eight-path snapshot: `c15cf8e7a3e0`. Review TASK-20261029-58d7a7: `844cbdf419c4`; completed release: `3e0b34465382`. Root claimed ledger TASK-20261029-bd67d5 at `f973a55dc291`, epoch 1. Exact hashes and parsed native receipt are in review-inputs.json.

The reviewer sealed precode-derivation.md with an observed 19:56:55 UTC clock and hash, then observed 19:57:00 UTC before source access. These are attested read-order observations, not independent timestamps or separate pre-code archival custody. Exact first-byte read time is unknown. Division-free checking and implementation-specific cost refinements are explicitly post-code. One reviewer is not a quorum or a line-by-line blind audit of producer prose. Native gpt-6.1-sol/xhigh was requested without provider authentication or an exposed opaque UUID. The reviewer records no post-dispatch exchanges.

The unchanged native checker returned zero at 2026-10-29T20:01:23.964499+00:00. Its exact parsed content and original receipt SHA256 are embedded in the manifest; parsed content is not described as identical JSON serialization. All producer/review/source hashes were compared successfully. No source, receipt or producer proof-status file is rewritten.

Root reported an additional S3 checkpoint push failed because the required ecc-challenges-new profile/upload token was unavailable and AWS profiles were empty. No upload, bypass or retry followed. Git publication is the recorded source/review custody; there is no verified new S3 copy. This infrastructure limitation asserts nothing about the mathematics.

Root must render and inspect every PDF page, fill actual rendering/custody fields, and verify the exact nine-path ledger archive. The manifest does not certify its own future archive. Existing diagram.svg, review-summary.svg and grouped-review-summary.svg in the regularity study were inspected and left unchanged: they are historical records of separate prior proof/review scopes. The present diagram supplements them without changing a shared frontier or scientific status.

Primary-source boundaries and record anchors

Salizzoni, Definitions 2.4–2.5, Propositions 3.10/3.13: arXiv:2304.13485. The specified mutant-closure upper bound is not actual algorithm necessity. Grouped raw s=2 retains input degree four and does not meet Proposition 3.13's input-degree premise.

RFC 9380, Appendix I and F.2.1.1: standard square-root and sqrt-ratio mechanisms, not novelty or reproduced performance. Coordinator rechecked the relevant primary text; source hashes are unknown.

Producer: cost-bridge.md Sections 1–5, reference.py, check_reference.py, test_reference.py and unit-test-results.json. Independent review: TASK-20261029-58d7a7/precode-derivation.md, blind-review.md and attestation.yaml. Prior algebra: DEC-20261029-ce5c57 and DEC-20261029-0857a9. Full identities, limitations, graph audit and future archive union: review-inputs.json.